zadanie 1.

$P(X=k)=e^{-\lambda}\frac{\lambda^k}{k!}$

X + Y = Z $\quad P(Y=k)=e^{-\mu}\frac{\mu^k}{k!}$

$$P(X+Y=k)=\sum_{i=0}^{k}P(X=i,Y=k-i)=\sum_{i=0}^{k}e^{-\lambda}\frac{\lambda^i}{i!}\cdot e^{-\mu}\frac{\mu^{k-i}}{(k-i)!}=e^{-(\lambda+\mu)}\frac{1}{k!}\sum_{i=0}^{k}\binom{k}{i}\lambda^i\mu^{k-i}=e^{-(\lambda+\mu)}\frac{(\lambda+\mu)^k}{k!}$$

$$P(Z=k)=e^{-(\lambda+\mu)}\frac{(\lambda+\mu)^k}{k!}$$

zadanie 2.

f(N) = $\frac{1}{N}$ - $\frac{1}{N+1}$, gdzie (N = 1, 2,…)

Aby funkcja określała rozkład zmiennej losowej dyskretnej muszą być spełnione warunki:

P(X = x_i) ≥ 0 oraz $\sum_{N=0}^{\infty}\left(\frac{1}{N}-\frac{1}{N+1}\right)=1$.

Pierwszy warunek jest spełniony, ponieważ ciąg jest większy lub równy 0.

Aby sprawdzić drug warunek musimy zbadać zbieżność szeregu.

$$S_n=\left(1-\tfrac{1}{2}\right)+\left(\tfrac{1}{2}-\tfrac{1}{3}\right)+\left(\tfrac{1}{3}-\tfrac{1}{4}\right)+\cdots+\left(\tfrac{1}{N}-\tfrac{1}{N+1}\right)=1-\tfrac{1}{N+1}$$

Dla N dążącego do nieskończoności granica szeregu wynosi 1, więc warunek jest spełniony.

Funkcja określa rozkład zmiennej losowej dyskretnej.

P(N ≥ 3) = 1 – P (N < 3) = 1 - [P(N = 1) + P(N = 2)] = 1 - $\frac{2}{3}$ = $\frac{1}{3}$

zadanie 3.

P(X = k) = p $(1-p)^{k-1}$, gdzie k = 1, 2, 3… oraz 0<p≤1

$$\sum_{k=1}^{n}q^{k-1}=\frac{1-q^n}{1-q}\text{, więc }\sum_{k=1}^{\infty}q^k=\frac{1}{1-q}$$

$$\sum_{k=1}^{\infty}kq^{k-1}=\frac{1}{(1-q)^2}=\frac{1}{p^2}$$

wartość oczekiwana:

$$E(X)=\sum_{k=1}^{\infty}k\cdot P(X=k)=\sum_{k=1}^{\infty}pk(1-p)^{k-1}=p\sum_{k=1}^{\infty}k\cdot q^{k-1}=p\cdot\frac{1}{p^2}=\frac{1}{p}$$, gdzie q = 1 - p

wariancja:

$D^2(X) = E(X^2) - (E(X))^2$

$$E(X^2)=\sum_{k=1}^{\infty}k^2q^{k-1}p=p\sum_{k=1}^{\infty}k^2q^{k-1}$$

$$[\sum_{k=1}^{\infty}kq^{k-1}]'=[\frac{1}{(1-q)^2}]'$$

$$\sum_{k=2}^{\infty}k(k-1)q^{k-2}=\frac{2}{(1-q)^3}$$

$$\sum_{k=1}^{\infty}k^2q^{k-1}+\sum_{k=1}^{\infty}kq^{k-1}=\frac{2}{(1-q)^3}$$

$$\sum_{k=1}^{\infty}k^2q^{k-1}+\frac{1}{(1-q)^2}=\frac{2}{(1-q)^3}$$

$$\sum_{k=1}^{\infty}k^2q^{k-1}=\frac{2}{(1-q)^3}-\frac{1}{(1-q)^2}=\frac{1+q}{(1-q)^3}$$

$$E(X^2)=\sum_{k=1}^{\infty}k^2q^{k-1}p=p\sum_{k=1}^{\infty}k^2q^{k-1}=(1-q)\frac{1+q}{(1-q)^3}=\frac{1+q}{(1-q)^2}$$

$$D^2(x)=\frac{1+q}{(1-q)^2}-\left(\frac{1}{1-q}\right)^2=\frac{1-p}{p^2}$$

zadanie 4.

Współczynnik asymetrii: ,znane: $\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{\lambda}$ $E(X) = \lambda$ $D^2(X) = \lambda$ $E(X^2) = \lambda^2 + \lambda$

$D^3(X) = E[(X-E(X))^3] = E(X^3) - 3E(X^2)E(X) + 3E(X)E^2(X) - E^3(X)$

$$E(X^3) = \sum_{k=0}^{\infty} k^3 e^{-\lambda}\frac{\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty} k^3\frac{\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty} k^2\frac{\lambda^k}{(k-1)!} = |\, r = k-1\,| = e^{-\lambda}\sum_{r=0}^{\infty}(r+1)^2\frac{\lambda^{r+1}}{t!} =$$

$$= \lambda e^{-\lambda}\sum_{r=0}^{\infty}(r^2+2r+1)\frac{\lambda^r}{r!} = \lambda e^{-\lambda}[\sum_{r=0}^{\infty} r^2\frac{\lambda^r}{r!} + 2\sum_{r=0}^{\infty} r\frac{\lambda^r}{r!} + \sum_{r=0}^{\infty}\frac{\lambda^r}{r!}] =$$

$$= \lambda e^{-\lambda}[\sum_{r=0}^{\infty} r\frac{\lambda^r}{(r-1)!} + 2\sum_{r=0}^{\infty}\frac{\lambda^r}{(r-1)!} + e^{\lambda}] = |l = r-1| = \lambda e^{-\lambda}[\sum_{l=0}^{\infty}(l+1)\frac{\lambda^{l+1}}{l!} + 2\sum_{l=0}^{\infty}\frac{\lambda^{l+1}}{l!} + e^{\lambda}] =$$

$$= \lambda e^{-\lambda}[\lambda(\sum_{l=0}^{\infty} l\frac{\lambda^l}{l!} + e^{\lambda}) + 2\lambda e^{\lambda} + e^{\lambda}] = \lambda e^{-\lambda}[\lambda(\sum_{l=0}^{\infty}\frac{\lambda^l}{(l-1)!} + e^{\lambda}) + 2\lambda e^{\lambda} + e^{\lambda}] = |\, h = l-1\,| =$$

$$= \lambda e^{-\lambda}[\lambda(\sum_{h=0}^{\infty}\frac{\lambda^{h+1}}{h!} + e^{\lambda}) + 2\lambda e^{\lambda} + e^{\lambda}] = \lambda e^{-\lambda}[\lambda(\lambda e^{\lambda} + e^{\lambda}) + 2\lambda e^{\lambda} + e^{\lambda}] =$$

$D^3(X) = \lambda^3 + 3\lambda^2 + \lambda - 3\lambda^3 - 3\lambda^2 + 3\lambda^3 - \lambda^3 = \lambda \rightarrow \gamma_1 = \frac{\lambda}{\lambda^{\frac{3}{2}}} = \lambda^{-\frac{1}{2}}$

Współczynnik skupienia:

$E^4(X) = E[(X-E(X))^4] = E(X^4) - 4E(X^3)E(X) + 6E(X^2)E^2(X) - E(X)E^3(X) + E^4(X)$

$$E(X^4) = \sum_{k=0}^{\infty} k^4 e^{-\lambda}\frac{\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty} k^4\frac{\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty} k^3\frac{\lambda^k}{(k-1)!} = |r = k-1| =$$

$$e^{-\lambda}\sum_{r=0}^{\infty}(r+1)^3\frac{\lambda^{r+1}}{r!} = \lambda e^{-\lambda}\sum_{r=0}^{\infty}(r^3+3r^2+3r+1)\frac{\lambda^r}{r!} =$$

$$= \lambda e^{-\lambda}[\sum_{r=0}^{\infty} r^3\frac{\lambda^r}{r!} + 3\sum_{r=0}^{\infty} r^2\frac{\lambda^r}{r!} + 3\sum_{r=0}^{\infty} r\frac{\lambda^r}{r!} + \sum_{r=0}^{\infty}\frac{\lambda^r}{r!}] =$$

$$= \lambda e^{-\lambda}[\sum_{r=0}^{\infty} r^2\frac{\lambda^r}{(r-1)!} + 3\sum_{r=0}^{\infty} r\frac{\lambda^r}{(r-1)!} + 3\sum_{r=0}^{\infty}\frac{\lambda^r}{r-1!} + e^{\lambda}] = \quad |l = r-1|$$

$$= \lambda e^{-\lambda}[\sum_{r=0}^{\infty}(l+1)^2\frac{\lambda^{l+1}}{l!} + 3\sum_{r=0}^{\infty}(l+1)\frac{\lambda^{l+1}}{l!} + 3\sum_{r=0}^{\infty}\frac{\lambda^{l+1}}{l!} + e^{\lambda}] =$$

$$= \lambda e^{-\lambda}[\lambda\sum_{l=0}^{\infty}(l^2+2l+1)\frac{\lambda^l}{l!} + 3\lambda(\sum_{l=0}^{\infty} l\frac{\lambda^l}{l!} + e^{\lambda}) + 3\lambda e^{\lambda} + e^{\lambda}] =$$

$$= \lambda e^{-\lambda}[\lambda(\sum_{l=0}^{\infty} l^2\frac{\lambda^l}{l!} + 2\sum_{l=0}^{\infty} l\frac{\lambda^l}{l!} + \sum_{l=0}^{\infty}\frac{\lambda^l}{l!}) + 3\lambda(\sum_{l=0}^{\infty} l\frac{\lambda^l}{l!} + e^{\lambda}) + 3\lambda e^{\lambda} + e^{\lambda}] =$$

$$= \lambda e^{-\lambda}[\lambda(\sum_{l=0}^{\infty} l\frac{\lambda^l}{(l-1)!} + 2\sum_{l=0}^{\infty}\frac{\lambda^l}{(l-1)!} + e^{\lambda}) + 3\lambda(\sum_{l=0}^{\infty}\frac{\lambda^l}{(l-1)!} + e^{\lambda}) + 3\lambda e^{\lambda} + e^{\lambda}] = \quad |h = l-1|$$

$$= \lambda e^{-\lambda}[\lambda(\sum_{h=0}^{\infty}(h+1)\frac{\lambda^{h+1}}{h!} + 2\sum_{l=0}^{\infty}\frac{\lambda^{h+1}}{h!} + e^{\lambda}) + 3\lambda(\sum_{h=0}^{\infty}\frac{\lambda^{h+1}}{h!} + e^{\lambda}) + 3\lambda e^{\lambda} + e^{\lambda}] =$$

$$= \lambda e^{-\lambda}[\lambda(\lambda(\sum_{h=0}^{\infty}\frac{\lambda^h}{(h-1)!} + e^{\lambda}) + 2\lambda e^{\lambda} + e^{\lambda}) + 3\lambda(\lambda e^{\lambda} + e^{\lambda}) + 3\lambda e^{\lambda} + e^{\lambda}] = \quad |c = h-1|$$

$$= \lambda e^{-\lambda}[\lambda(\lambda(\sum_{c=0}^{\infty} \frac{\lambda^{c+1}}{c!} + e^{\lambda}) + 2\lambda e^{\lambda} + e^{\lambda}) + 3\lambda^2 e^{\lambda} + 3\lambda e^{\lambda} + 3\lambda e^{\lambda} + e^{\lambda}] =$$

$$= \lambda e^{-\lambda}[\lambda(\lambda(\lambda e^{\lambda} + e^{\lambda}) + 2\lambda e^{\lambda} + e^{\lambda}) + 3\lambda^2 e^{\lambda} + 6\lambda e^{\lambda} + e^{\lambda}] =$$

$$= \lambda e^{-\lambda}[\lambda^3 e^{\lambda} + \lambda^2 e^{\lambda} + 2\lambda^2 e^{\lambda} + \lambda e^{\lambda} + 3\lambda^2 e^{\lambda} + 6\lambda e^{\lambda} + e^{\lambda}] =$$

$$= \lambda e^{-\lambda}[\lambda^3 e^{\lambda} + 6\lambda^2 e^{\lambda} + 7\lambda e^{\lambda} + e^{\lambda}] = \lambda^4 + 6\lambda^3 + 7\lambda^2 + \lambda$$

$$D^4(X) = \lambda^4 + 6\lambda^3 + 7\lambda^2 + \lambda - 4\lambda(\lambda^3 + 3\lambda^2 + \lambda) + 6\lambda^2(\lambda^2 + \lambda) - 4\lambda^4 + \lambda^4 =$$

$$= -2\lambda^4 + 6\lambda^3 + 7\lambda^2 + \lambda - 4\lambda^4 - 12\lambda^3 - 4\lambda^2 + 6\lambda^4 + 6\lambda^3 = 3\lambda^2 + \lambda$$

$$\beta = \frac{D^4(X)}{(D^2(X))^2} - 3 = \frac{3\lambda^2+\lambda-3\lambda^2}{\lambda^2} = \frac{1}{\lambda} = \lambda^{-1}$$

zadanie 5.

n = 500 p = 0,002

$P(X \le 2) = P(X = 0) + P(X = 1) + P(X = 2)$,

$$P(X = 0) = \binom{500}{0} \cdot 0{,}002^0 \cdot 0{,}998^{500}$$

$$P(X = 1) = \binom{500}{1} \cdot 0{,}002^1 \cdot 0{,}998^{499}$$

$$P(X = 2) = \binom{500}{2} \cdot 0{,}002^2 \cdot 0{,}998^{498}$$

$$P(X \le 2) = \binom{500}{0} \cdot 0{,}002^0 \cdot 0{,}998^{500} + \binom{500}{1} \cdot 0{,}002^1 \cdot 0{,}998^{499} + \binom{500}{2} \cdot 0{,}002^0 \cdot 0{,}998^{498}$$

zadanie 6.

rozkład geometryczny: $P(X=k) = p(1-p)^{k-1}$

$P(X \le m_e) = \frac{1}{2} = P(X \ge m_e)$

$P(X \ge m_e) = 1 - P(X < m_e)$

$\frac{1}{2} = 1 - P(X < m_e)$

$P(X < m_e) = \frac{1}{2}$

$$\sum_{k=1}^{me} p(1-p)^{k-1} = \frac{1}{2}$$

$(1 - p) = q$

$$p\sum_{k=1}^{m_e} q^{k-1} = \frac{1}{2}$$

$$p \cdot \frac{1-q^{m_e}}{1-q} = \frac{1}{2}$$

$1 - q^{me} = \frac{1}{2}$

$q^{me} = \frac{1}{2}$

$e^{me \ln q} = e^{-\ln 2}$

$e^{me \ln(1-p)} = e^{-\ln 2}$

$m_e \ln(1-p) = -\ln 2$

$$m_e = \frac{-\ln 2}{\ln(1-p)}$$

zadanie 7.

a) Udowodnić $E(\overline{X}) = E(X)$

$$E(\overline{X}) = E\left(\frac{1}{n}\sum x_i\right) = \frac{1}{n}E\left(\sum x_i\right) = \frac{1}{n}\left(E(x_1)+E(x_2)+\ldots+E(x_n)\right) = \frac{1}{n}\cdot n\,E(X) = E(X)$$

b) Udowodnić $D^2(\overline{X}) = \frac{D^2(X)}{n}$

$$D^2(\overline{X}) = D^2\left(\frac{1}{n}\sum x_i\right) = \frac{1}{n^2}D^2\left(\sum x_i\right) = \frac{1}{n^2}\left(D^2(x_1)+D^2(x_2)+\ldots+D^2(x_n)\right) = \frac{D^2(X)}{n}$$

zadanie 8.

$n = 16$, $\sigma = 10$, $D^2(X) = 100$, gdzie: **σ** to odchylenie standardowe średniej arytmetycznej,
n to ilość zmiennych niezależnych od siebie
$D^2(\overline{X})$ to wariancja zmiennej losowej $\overline{X}$

$$D^2(\overline{X}) = \frac{100}{16} = \frac{25}{4}$$

$$\sigma = \sqrt{\frac{25}{4}} = \frac{5}{2}$$